CLOUD & DEVOPS

Auto-scaling Visual Guide

HPA, VPA, cluster autoscaler — right-size everything, automatically.

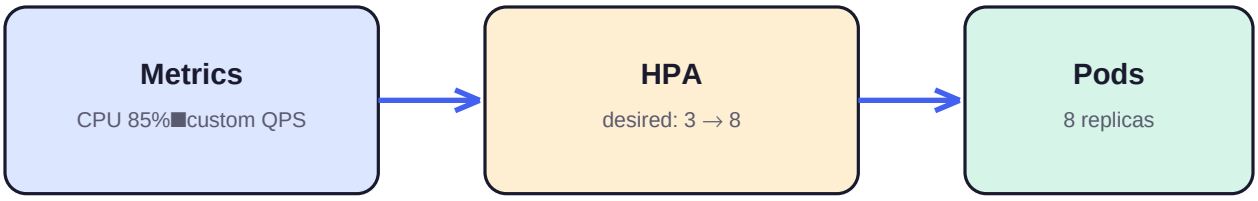
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

HPA: Scale Pods on Metrics

Horizontal Pod Autoscaler adds replicas when load rises.



Control loop: every 15s, HPA computes desiredReplicas = ceil(current × (metric/target)).

HPA needs resource requests set — without them, CPU% has no denominator and HPA can't compute.

STEP 2/6

Choosing the Right Metric

CPU is easy; custom metrics are accurate.

CPU utilization

simple, laggy — bad for spiky or queue-driven work

Memory

rarely useful — memory doesn't shed load when you add pods

Custom: requests/sec

scale on what users feel — needs metrics adapter

Custom: queue depth

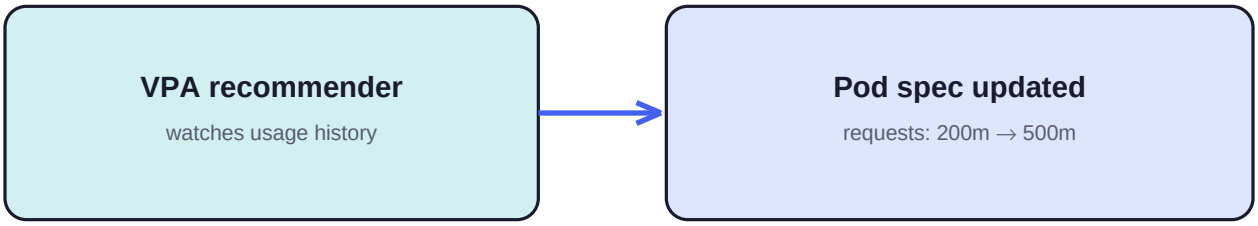
workers scale with backlog — the honest signal

Multiple metrics: HPA takes the MAX of all computed replica counts — the most demanding signal wins.

STEP 3/6

VPA: Right-Size the Containers

Vertical Pod Autoscaler adjusts requests/limits per pod.



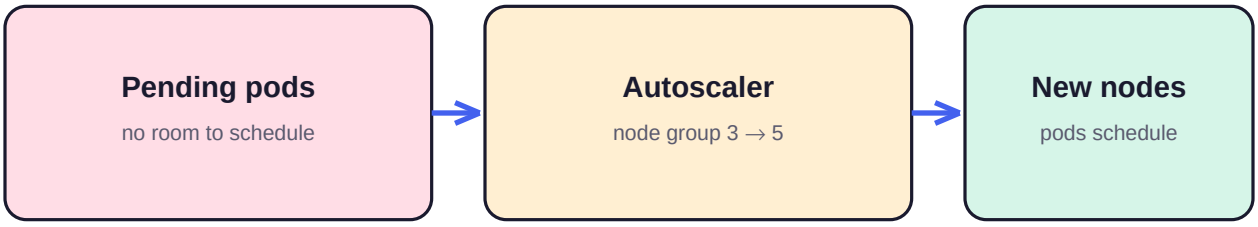
Modes: Off (recommend only) → Initial → Auto (restarts pods to apply).

Never run HPA and VPA on the SAME metric — they fight. VPA for sizing, HPA for replica count, different signals.

STEP 4/6

Cluster Autoscaler: Scale the Nodes

Pods pending? Add nodes. Nodes empty? Remove them.



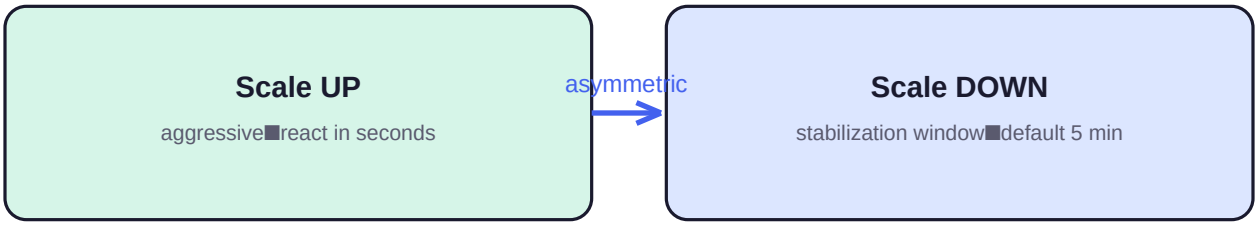
Scale-down: evicts pods from underutilized nodes (respects PodDisruptionBudgets), then deletes the node.

Node startup takes minutes — cluster autoscaler reacts to sustained pressure, not spikes. HPA handles the spikes.

STEP 5/6

Cooldowns & Stabilization

Prevent flapping: scale up fast, scale down slow.



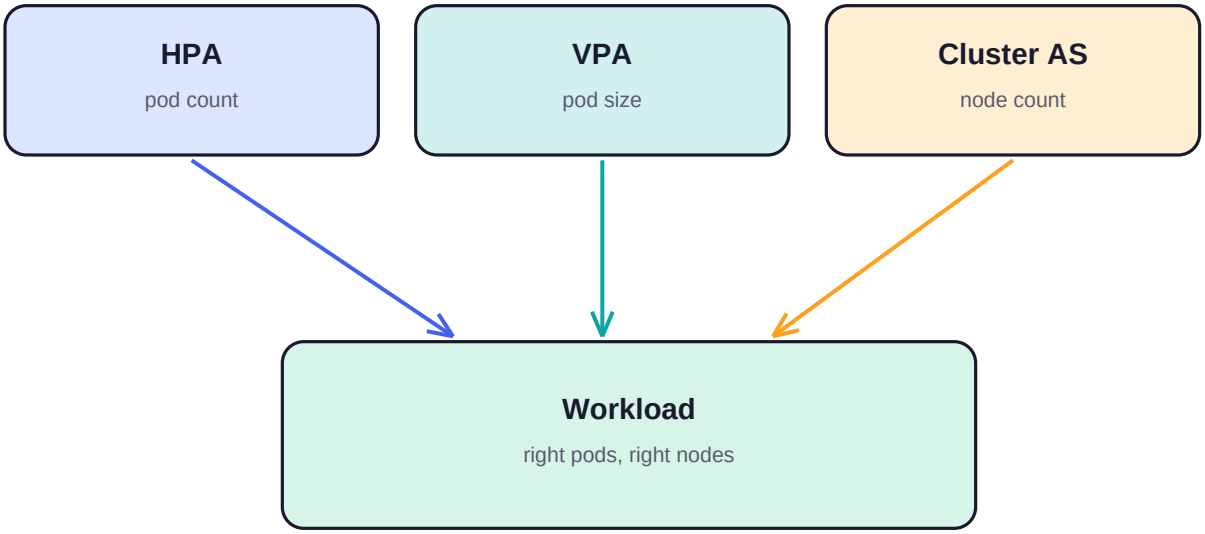
Flapping: 3→8→3→8 replicas every minute — churns connections and confuses the cluster autoscaler.

- scaleDown.stabilizationWindowSeconds: ignore brief dips before shrinking.
- scaleUp can also be rate-limited (policies) to avoid thundering herds.
- Predictive/scheduled scaling: pre-warm before known peaks (Black Friday 9am).

STEP 6/6

The Full Stack, Together

Three scalers, three different jobs, one system.



HPA reacts in seconds, cluster autoscaler in minutes — size node headroom for the gap.

STEP ✓/✓

Key Takeaways

Auto-scaling — Remember

- HPA scales pod COUNT on metrics — set resource requests or it can't compute.
- Scale on honest signals: request rate or queue depth beat raw CPU.
- VPA right-sizes containers; never let it fight HPA on the same metric.
- Cluster autoscaler adds/removes NODES for pending pods and empty nodes.
- Scale up fast, scale down slow — stabilization windows stop flapping.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev